Code No: 117CD **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
DATA WAREHOUSING AND DATA MINING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions

JNTUH Used papers 2023

PART – A

(25 Marks)

1.a)	Differentiate between base cuboid and apex cuboid.	[2]
b)	What are the differences between fact table and dimension table?	[3]
c)	What do you mean by numerosity reduction?	[2]
d)	What is Data Discretization?	[3]
e)	What is the purpose of Apriori Algorithm?	[2]
f)	Define support and confidence in Association rule mining.	[3]
g)	What is meant by classification? What are the applications of classification?	[2]
h)	State and explain Bayes theorem.	[3]
i)	What are the difference between K-Means and K-Medoids Algorithms?	[2]
j)	Define the centroid of the cluster.	[3]

PART – B

(50 Marks)

2.a)	Draw and explain the three tier data warehouse architecture.	
b)	Differentiate OLTP and OLAP with features.	[5+5]
	OR	
3.a)	Define metadata and explain the types of metadata.	
b)	Compare the types of OLAP Servers.	[5+5]
4.a)	What is KDD? Explain about data mining as a step in the process of knowledge discovery.	
b)	Discuss about Data Mining task primitives with examples.	[5+5]
	OR	
5.a)	Discuss the issues to be considered during data integration.	
b)	How can we handle missing values? Illustrate with an example.	[5+5]
6.a)	Discuss the importance of Association Rule Mining.	
b)	Write a note on Maximal Frequent Item Set and Closed Frequent Item Set.	[5+5]
	OR	
7.	Can we design a method that mines the complete set of frequent item sets without candidate generation? If yes, explain with an example.	[10]

8.a) What is Bayesian belief network? Explain in detail.
b) Explain tree pruning. [5+5]

OR

9. Write and explain about Classification by Back propagation Algorithm. [10]

10.a) What is Clustering? What are different types of clustering?
b) Give a brief note on PAM Algorithm. [5+5]

OR

11.a) What is outlier detection? Explain distance based outlier detection
b) What is the drawback of k-means algorithm? How can we modify the algorithm to diminish that problem? [5+5]

---ooOoo---

JNTUH Used papers 2023